

Wave 2 Peergos apps – high-level description

Seven installable Peergos apps close the UI gaps after hello-cube, IT wine, and signatures. Each app is its own git repository. They store state in the Peergos Cryptree via `/peergos-api/v0/data`. They never open NATS. Connectors on the desktop host watch `outbox/` and publish JetStream subjects registered in `verae-nats-bus`.

App	Repo	Specialist	What the user sees
Cube Browser	verae-app-cube-browser	JS + integrity	List cubes
Share Wizard	verae-app-share-wizard	Frontend + integrity	Mode
TS Inbox	verae-app-timestamp-inbox	Timestamp + JS	Local/Share
ISO Export	verae-app-iso-export	Storage + wine UI	Queue `vera`
ITAD Board	verae-app-itad-board	ITAD SME + JS	pickup→wipe→sal
Sig Verify	verae-app-sig-verify	Crypto + JS	valid/revoked + `
Conn Status	verae-app-connector-status	Messaging + host	outbo

Plan: [PLAN-APPS-WAVE2.md](https://github.com/DataCube-Modules/Verae-in-Solution-Pieces) (also copied conceptually here).

Peergos-compliant means: valid `peergos-app.json`, `assets/index.html` no WebSocket/NATS in the iframe, `fetch` only to `/peergos-api` or relative files, unit tests green. Certified with `make certify` on 2026-08-16.

Wave 2 installation guide

Prerequisites

- Peergos account on <https://peergos.georgelambert.org> (or local daemon)
- Java 21 + `Peergos.jar` if using OrgTool
- Desktop-host connectors for outbox subjects (timestamp, append, exp

Install one app

From a clone of e.g. `verae-app-cube-browser`:

```
```bash
Web: upload peergos-app.json + assets/, then Install App
CLI:
java -cp Peergos.jar OrgTool install-app \
 https://peergos.georgelambert.org "$USER" "$PASS" cube-browser .
```
```

Or wait for gallery publish: Apps → Custom → Install.

Install all seven

```
```bash
for r in cube-browser share-wizard timestamp-inbox iso-export itad-bo
 echo "== $r =="
 # OrgTool install-app ... $r ~/research/verae-app-$r
done
```

```

...

Connector subjects to enable on the host

| Subject | App |
| --- | --- |
| `verae.ts.request` / receipts | timestamp-inbox (reads inbox) |
| `verae.cube.append` | itad-board |
| `verae.cube.export` | iso-export (new) |
| `verae.sig.verify` | sig-verify (new) |

`verae.cube.export` and `verae.sig.verify` were added to
`verae-nats-bus/schemas/subjects.json`. Host connectors must implement
them before those two apps do useful work beyond queueing files.

Verify

```bash
cd ~/research/verae-app-cube-browser && make certify
```

Repeat per repo. Expected: `peergos compliance: PASS`.

```

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## # Wave 2 developer reference

Each repo follows the same layout:

```

...
peergos-app.json
assets/index.html # sandbox UI
assets/sdk.js # browser VeraeSDK
src/peergos-sdk.mjs # ESM for tests
src/lib.mjs # specified functions
tests/lib.test.mjs
tests/certify-peergos.sh
docs/SPECS.md # function I/O + behavior
docs/INSTALL.md
docs/DEVELOPER.md
...

```

## ## Function index

```

Repo	Functions
cube-browser	`listCubes`, `listCubeObjects`, `extractHashes`
share-wizard	`buildSharePlan`, `writeSharePlan`, `planIncludes`
timestamp-inbox	`listInbox`, `parseReceipt`, `scopeRank`
iso-export	`enqueueExport`, `parseExportAck`
itad-board	`STAGES`, `nextStage`, `columnize`, `enqueueStage`
sig-verify	`listValid`, `listRevoked`, `isRevoked`, `enqueueVerify`
connector-status	`listOutbox`, `listInbox`, `jobStem`, `classifyJ

```

See each `docs/SPECS.md` for inputs, outputs, and internal behavior.

## ## Shared client

```
`createClient(fetch)`:
```

```
- `writeJSON(path, obj)` → PUT `/peergos-api/v0/data`+path
- `readJSON(path)` → GET JSON
- `list(path)` → `{files, subFolders}` (Peergos folder GET)
```

```
`makeEnvelope(subject, payload, reply_file)`
`enqueue(client, name, envelope)` → `/outbox/<name>.json`
```

## ## Compliance

```
`make certify` fails if assets contain `nats.connect`, `WebSocket`,
`wss://`, or absolute `http(s)` `fetch`. Manifest displayName ≤ 25,
description ≤ 100, schemaVersion 1, launchable true.
```

## # Wave 2 Peergos apps – plan

### ### Comments:

Independent installable Peergos apps that close UI gaps after hello-c  
ITAD / wine / signatures. Each app is its **\*\*own git repo\*\***. Apps neve  
open NATS; they write Peergos files. Connectors on `verae-desktop-hos  
watch `/.apps/<name>/data/outbox/` and publish JetStream.

## ## Integration (Peergos + files + NATS)

```
` ``
```

### Sandbox app

```
→ PUT/GET /peergos-api/v0/data/... (Cryptree files)
→ PUT /outbox/<job>.json {subject, payload, reply_fil
```

### Connector (host, not sandbox)

```
→ NATS JetStream verae.*
→ PUT /inbox/<reply>.json
```

```
App polls or lists /inbox/ (still no NATS)
```

```
` ``
```

| Subject           | Who writes outbox                          | Payload                         | Inbox reply  |
|-------------------|--------------------------------------------|---------------------------------|--------------|
| verae.ts.request  | timestamp-inbox (re-stamp), others already | h                               |              |
| verae.cube.append | itad-board                                 | block                           | optional ack |
| verae.cube.export | iso-export                                 | cube_id, format=iso, share_mode |              |
| verae.sig.verify  | sig-verify                                 | object_sha256, signature_id     | {ok          |

File roots (all under `/.apps/<app>/data/`):

```
- `/cubes/<id>/manifest.json` + objects
- `/outbox/*.json` pending jobs
- `/inbox/*.json` connector replies
- `/signatures/valid|revoked/*.json`
```

## ## Repos, specialists, functions

| Repo                      | Specialist              | Pattern | Functions         |
|---------------------------|-------------------------|---------|-------------------|
| verae-app-cube-browser    | JS + integrity          | A+C     | `listCubes`, `lis |
| verae-app-share-wizard    | Frontend + integrity    | A       | `validateShar     |
| verae-app-timestamp-inbox | Timestamp + JS          | A       | `listInbox`, `pa  |
| verae-app-iso-export      | Storage + wine frontend | A       | `enqueueExpo      |

```
| `verae-app-itad-board` | ITAD SME + JS | A | `STAGES`, `columnize`,
| `verae-app-sig-verify` | Crypto + JS | A | `listValid`, `listRevoke`
| `verae-app-connector-status` | Messaging + host | A | `listOutbox`,
```

## Peergos compliance (certify)

1. `peergos-app.json`: schemaVersion 1, displayName  $\leq 25$ , description
2. Published tree = `peergos-app.json` + `assets/` (`index.html` requ
3. Assets never mention `nats.connect`, `ws://`, `wss://`, or absolut
4. All `fetch` targets start with `/peergos-api/` or are relative.
5. Unit tests pass (`make test`).
6. Outbox envelopes use registered subjects only.

## Build / test / docs

Each repo: `src/\*.mjs` (specified functions)  $\rightarrow$  `tests/\*.test.mjs`  $\rightarrow$   
`assets/` UI  $\rightarrow$  `docs/{SPECS,INSTALL,DEVELOPER}.md`  $\rightarrow$  `make test` +  
`make certify`.

Handbook PDFs: `docs/published/` in this wave's handbook or  
`verae-peergos-app-handbook/docs/published/`.

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